Fermentation

Fermentation is the microbial conversion of starch and sugars into alcohol. It makes foods more nutritious and palatable.

Pickling

Pickling is a method of preserving food in an edible antimicrobial liquid. It is of two types: chemical pickling and fermentation pickling.

In chemical pickling, food is placed in an edible liquid that kills bacteria and other microorganisms. Eg. Vinegar, alcohol, vegetable oil (pickling agents). In fermentation pickling, bacteria in the liquid produce organic acid as preservation agent that produces lactic acid due to the presence of *Lactobacillus*.

Boiling

Boiling liquid food items kill all the microbes. Eg. Milk and Water.

Sugaring

Sugar is used to preserve fruits in an antimicrobial syrup with fruits such as apples, pears, peaches, plums or in a crystallized form, so that the product is stored in dry condition.

2. Modern techniques

Pasteurization

It is a process for preservation of liquid food. This method was invented by Louis Pasteur in 1862. Milk is preserved by this method. It is heated up to 70 °C to kill the bacteria and it is cooled to 10 °C to prevent the growth of remaining bacteria. Then milk is stored in sterilized bottles in cold places.

b. Food production

Probiotics

Probiotics are live food supplements used in yoghurt and other fermented milk products. Eg. *Lactobacillus acidophilus* and *Bifidobacterium bifidum*. These bacteria improve the microbial spectrum in the gut and thus contribute to the following effects.

- Decrease the risk of colon cancer
- Decrease cholesterol absorption
- Prevent diarrheal diseases by increasing immunity.

16.9 Harmful Microorganisms

A few microorganisms are harmful to plants, animals and humans. They cause diseases and hence they are called as pathogens. Pathogens enter into the body through cuts and wounds in the skin, mouth or nose and cause diseases. Viruses causing 'flu' are spread through air. When the patient sneezes droplets containing viruses spread in air and it gets entered to another person when he breathes. Some of the diseases caused by the microorganisms in plants, animals and humans are given in the tables below.



Citrus canker Potato blight disease

Figure 16.16 Disease in plants

Table 16.1 Diseases caused by microorganisms in plants

Plant diseases	Causative microorganisms	Mode of transmission	Symptoms	Preventive measures/ Treatment
Citrus canker	*Xanthomonas axonopodis* (Bacteria)	Air, water	Lesions on leaves, stems and fruit	Copper based bactericides can be used
Potato blight disease	*Phytophthora infestans* (Fungi)	Air	Brown lesions on the surface of tubers	Fungicides are used

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